1.

In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Result: In Cisco Packet Tracer, I configured one DHCP server with a starting IP address of 192.168.1.10 and connected two client PCs. Both PCs received different IP addresses automatically from the DHCP pool**.**

- **PC0: 192.168.1.10**
- **PC1: 192.168.1.11**

Conclusion: DHCP successfully assigned unique IP addresses to both PCs automatically, without requiring manual IP configuration.

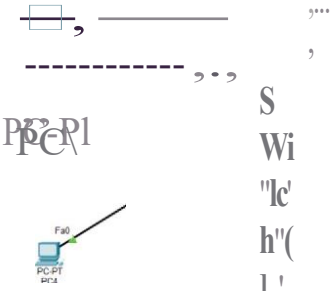
, Ckco P.ad::□ Tr*MM*•
C:',U□m,□h1(\0n□Om,□\I)Mlo.1.op\TOPS\CISCO\DHCP·DIIIS-WUI
SM<M ConF.gur.'IMn.plrt □ X
:Tlo!' Ed,! OphoM V, Tool□ □!.ion!: WiMow □p
i□□e•r□@□□□□□□□□m□□
□o_ci:::;.:EI/•• I li:il'-1





"P"C"O'

'''' 192161121 ''''





ii lii •• .1.= £\ □ 13 f' 1, ®-.□□□□ lii -i lii
] o_ cl ::LI! __/ mQ
.ip PCO -□
Logical

Physical Config Deskto **Programming** Attributes

IP Configuration

Interface FastEthernetO

IP Configuration

0 ☐HCP

0 Static

IPv4Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	192.168.2.3

IPv6 Configuration

0 Automatic

0 Static

IPv6Address	
Link Local Address	FE80::201:C7FF:FE4A:77☐
Default Gateway	
DNS Server	

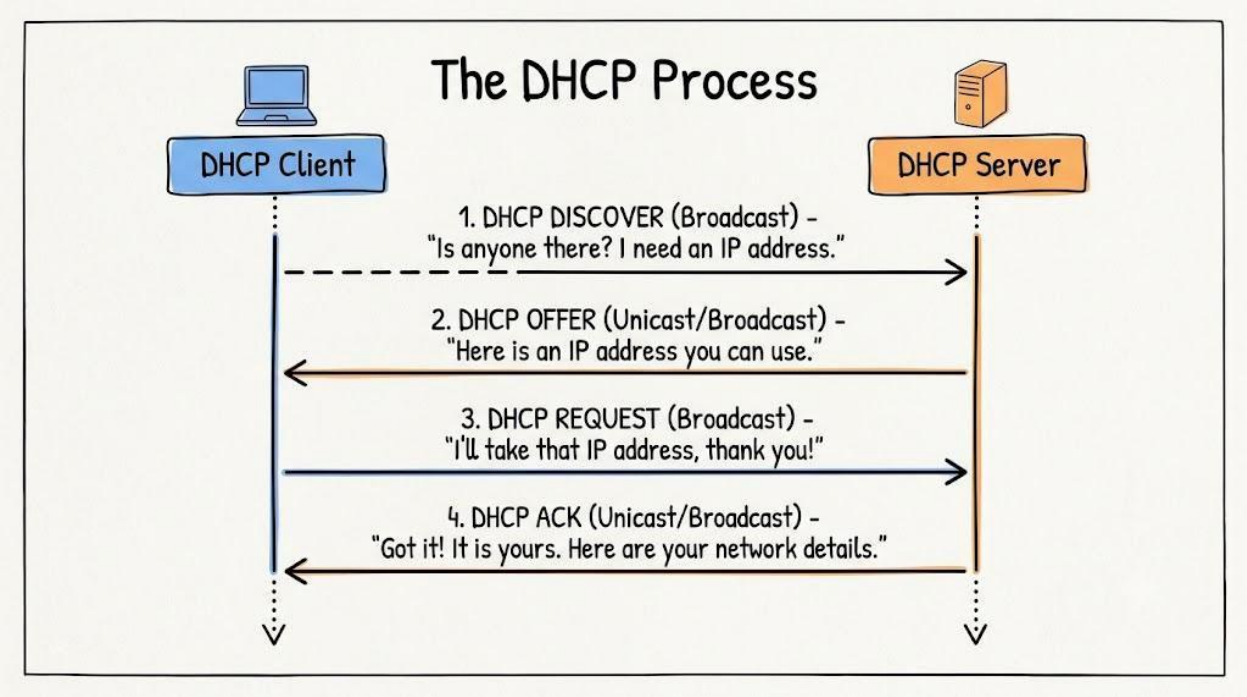
802.1X

0 Use 802.1X Security

Authentication	MD5
Username	
Password	

2.

Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.



Discover (Client Server): The client broadcasts a message asking if there is an IP address available for use.

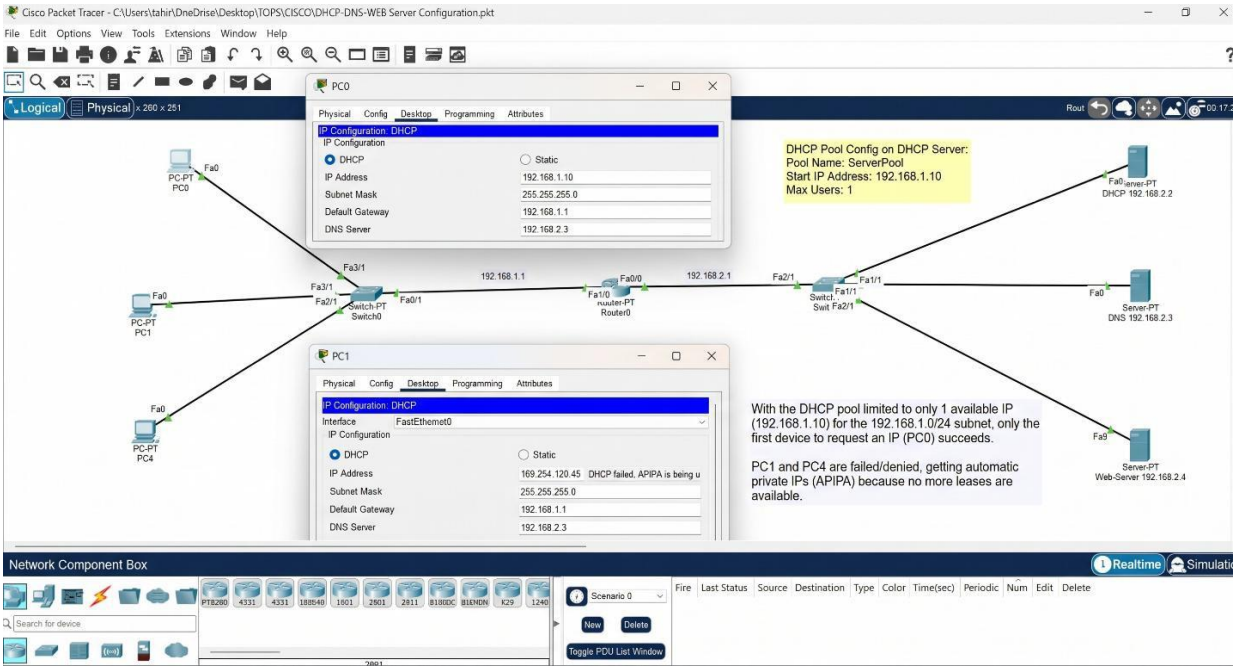
Offer (Server Client): A DHCP server responds back with an available IP address option.

Request (Client Server): The client broadcast-requests to accept that offered IP address.

Acknowledge (Server Client): The server confirms the reservation and sends the required network details (subnet mask, DNS, gateway) to finish the setup.

3.

Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



4.

Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

